

ISPyB status report

Rasmus Fogh, Global Phasing
MXCuBE/ISPyB Meeting, HZB Berlin, 10 June 2026

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- We have no beamlines and do not run a LIMS
 - Our software (Workflow, autoPROC, ...) runs at beamlines, takes inputs from beamline LIMS and sends outputs to beamline LIMS and viewers
 - Our users depend on getting their data and metadata out of synchrotrons
 - Our software provides added value (multi-sweep data sets, customised strategies, in-depth analysis) and we depend on optimal execution and full display of results going beyond the lowest common denominator
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- Our software and users work at multiple synchrotrons and would benefit from uniformity of LIMS access and result viewers
 - The increasing fragmentation of the synchrotron LIMS landscape works against standardisation, and forces people to access each synchrotron in a different way.
 - We work with synchrotrons on feedback and on how to store and present (our) complex metadata to users.
 - See Clemens Vonnrhein's talk tomorrow for details
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- Given the well-established impossibility of having uniform LIMS systems at different synchrotrons, we are working on a collaborative data model, API, and exchange format: MXLIMS for uniform access and interoperability
 - MXLIMS version 0.6.13 is stable and covers MX samples, shipping, and processing results
 - See my talk tomorrow for more on MXLIMS
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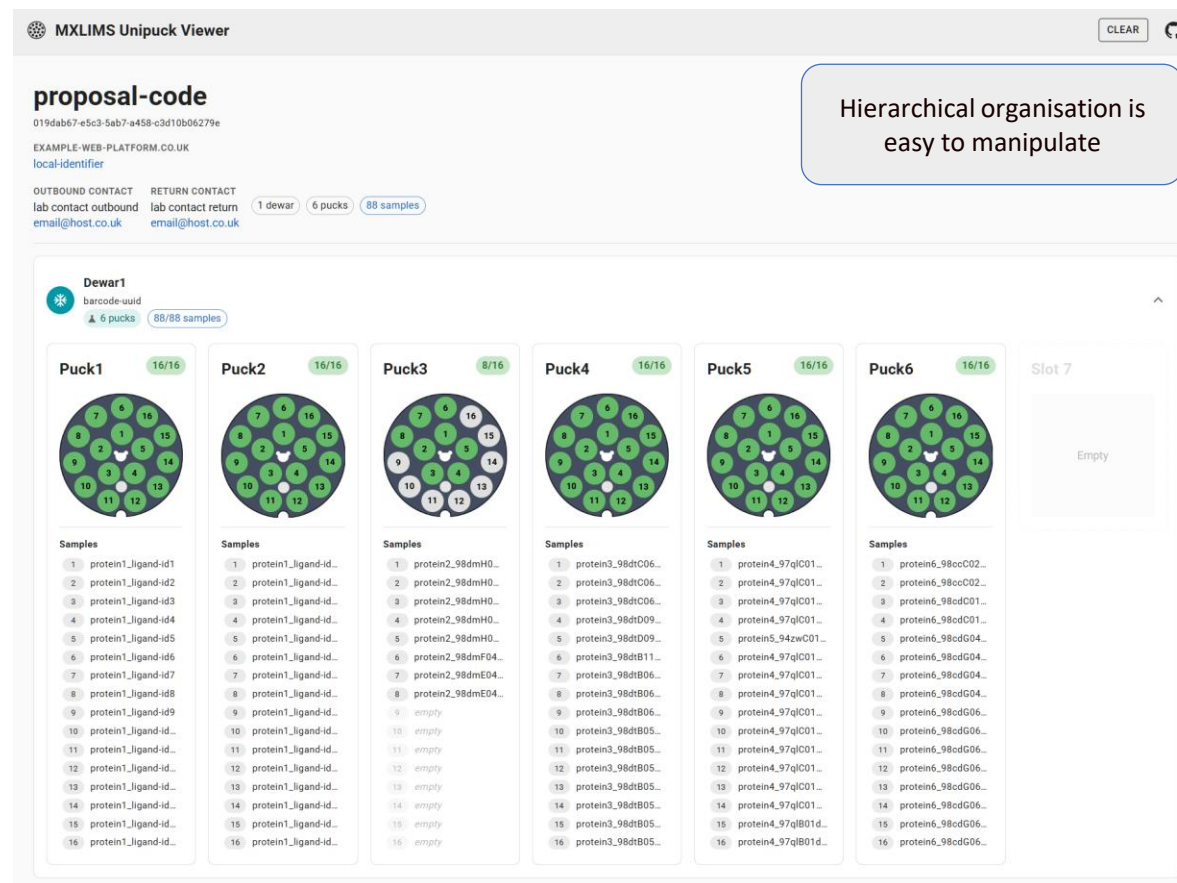
- A 'Coalition of the Willing' is currently testing the use of MXLIMS files for sample shipping
- Several sites have generated MXLIMS shipment messages:
Ed Daniel (IceBear), Dennis Stegmann (SLS-FFCS), Gustavo Lima (Astex)
- Guilherme Freitas (DLS) has created a shipment ingestion endpoint and has successfully ingested the test messages into the DLS test ISPyB database.
- We are working on generating and incorporating MXLIMS messages for returning experiment and processing results
- Gustavo Lima reports integrating MXLIMS with internal ASTEX databases.

The figure displays four panels of code snippets, likely from different repositories or projects, showing various code lines and their corresponding line numbers. The panels are labeled as follows:

- diamond**: Shows code lines 1 through 10, including comments like "The following code is for the diamond synchrotron" and "The following code is for the diamond synchrotron".
- ESRF**: Shows code lines 1 through 10, including comments like "The following code is for the ESRF synchrotron" and "The following code is for the ESRF synchrotron".
- MAXIV**: Shows code lines 1 through 10, including comments like "The following code is for the MAXIV synchrotron" and "The following code is for the MAXIV synchrotron".
- FEL**: Shows code lines 1 through 10, including comments like "The following code is for the FEL synchrotron" and "The following code is for the FEL synchrotron".

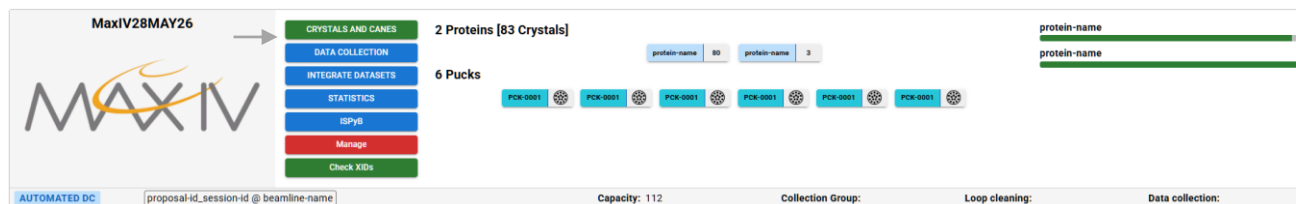
The code snippets are presented in a grid-like format, with each panel containing a list of code lines and their corresponding line numbers. The code is color-coded, with some lines highlighted in blue. The panels are arranged horizontally, with the labels "diamond", "ESRF", "MAXIV", and "FEL" positioned above their respective code snippets.

Updates could be easier to follow
(or be invisible, in some cases)



- Quick view of the MXLIMS JSON declaration.
 - Currently expecting Unipucks and CX-100 dewars.

Our implementation



MaxIV28MAY26

CRYSTALS AND CANES 2 Proteins [83 Crystals]

DATA COLLECTION

INTEGRATE DATASETS

STATISTICS

ISPyB

Manage

Check XIDs

6 Pucks

protein-name 80 protein-name 3

protein-name

protein-name

AUTOMATED DC proposal-id_session-id @ beamline-name Capacity: 112 Collection Group: Loop cleaning: Data collection:

ISPyB definitions for DATA-COLLECTION-SESSION

Devars DEWAR-0001

Pucks on DEWAR-0001 PCK-0001, PCK-0002, PCK-...

AUTOSIZE SAVE LOAD CHECK ISPyB OPEN DC MONITOR REGISTER ON ISPyB PROFORMA SHIPPING LABELS Search...

container position	protein acronym	sample acronym	experimentType	aimed Resolution	required Resolution	beam diameter	total rot. angle	min osc. angle	comments	energy	transmission	exposure time	oscillation range
1	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
2	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
3	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
4	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
5	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
6	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
7	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
8	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
9	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
10	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1
11	acronym	acronym_crystal...	Default	1.5	2	50	360	0.1		17000	100	0.002	0.1

Print

Export as CSV (selected)

Export as CSV (all rows)

Download as Excel

Export MXLIMS

- Users predefine data collection parameters for the project.
- We map our database schema into a common DataGrid object.
- Using DataGrid built-in methods, we can export the data in the required format with a single parser.
 - Includes export to MXLIMS format (tested at Diamond)

- Thanks to the many, many people who have helped making the MXLIMS model and participated in the MXLIMS working group, the MXCuBE automation WG, or contributed information, discussions and feedback.
- Particular thanks to the Coalition of the Willing:
Ed Daniel (Icebear), May Sharpe and Dennis Stegmann (SLS-FFCS),
Guilherme Freitas, James Hall, and Jose Brandao-Neto (Diamond),
Ezequiel Panepucci (MAX IV), and Gustavo Lima (Astex)